

Announcements

- Assignment coming up

Today

- More continuous time
- Exponential distribution
- Poisson process
- Poisson distribution

When does the individual die?

- Discrete approximation to continuous time
 - small interval of time, Δt
 - constant probability of death, p_{death}
- Set time interval very small
 - $\Delta t \rightarrow 0$

When does the individual die?

Algorithm

set $m, \Delta t, p_{\text{death}} = m\Delta t$

set state = alive, $t = 0$

while state = alive

$t = t + \Delta t$

 if death occurs given Bernoulli(p_{death})

 state = dead

print t (time to death)

Bernoulli process

Poisson process

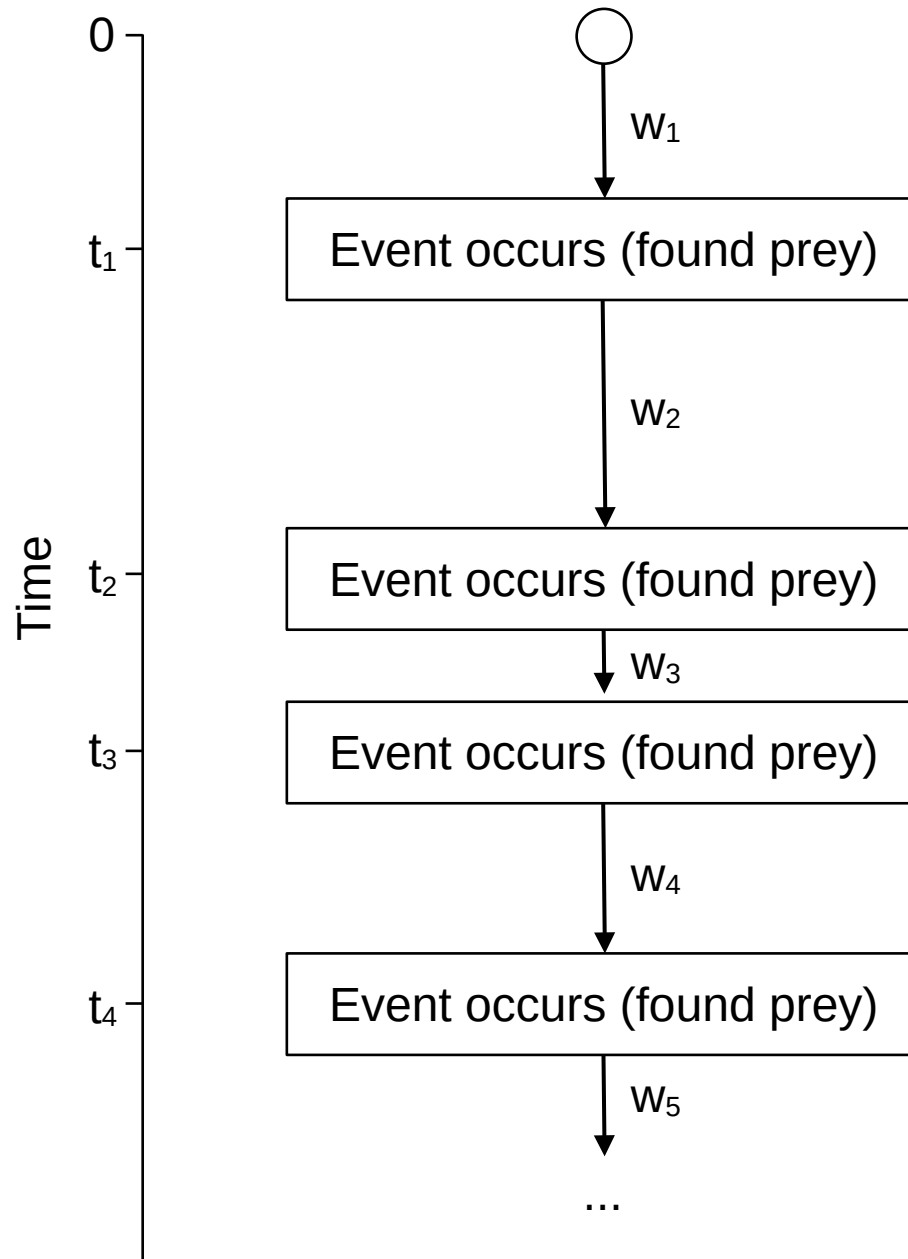
- A continuous-time process
- Waiting time between events is an exponential random variable
- e.g. individual searching for prey
 - search success rate = s

Poisson process

Event
sequence
diagram

t: event time
w: waiting time

$$w \sim \text{Exp}(s)$$



Poisson process

Individual searching for prey items

Algorithm

set s , $t_{\max}$

set $t = 0$

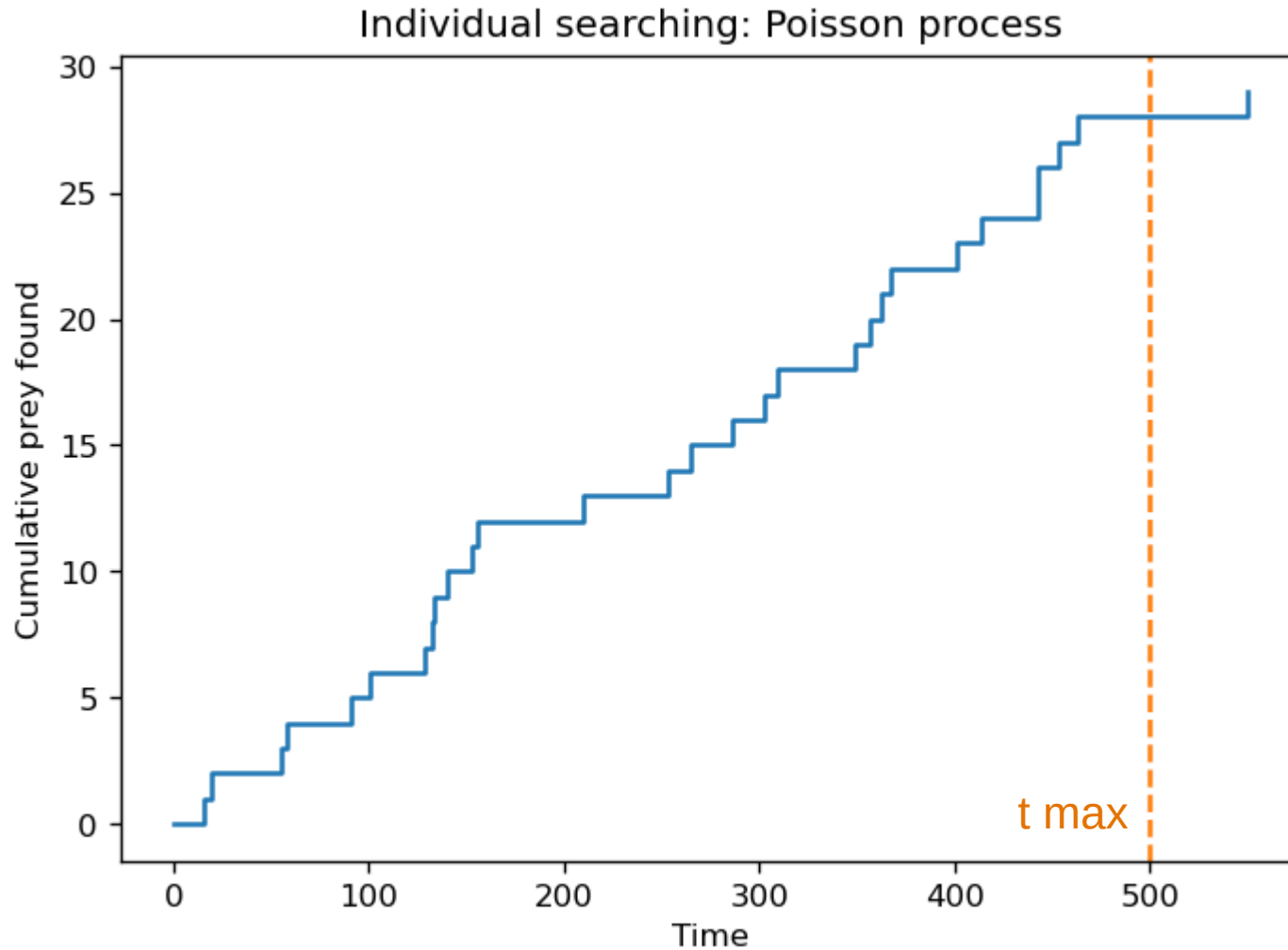
while $t < t_{\max}$

 draw waiting time w from $\text{Exp}(s)$

$t = t + w$

 prey item is found

Poisson process



Poisson distribution

- Emerges as the **sum of events** in a Poisson process
- Number of events that occur in a fixed time $t_{\max}$
- How many events occur before the sum of exponentially distributed waiting times exceeds $t_{\max}$?

Poisson distribution

Individual searching for prey items

Number of prey found before $t_{\max}$

Algorithm

set s , $t_{\max}$

set $t = 0$, $\text{prey} = 0$

while $t < t_{\max}$

 draw waiting time w from $\text{Exp}(s)$

$t = t + w$

$\text{prey} = \text{prey} + 1$ (prey item found)

return $\text{prey} - 1$

Poisson distribution

